THE CENTRALIZER OF A TENSOR PRODUCT OF SUBALGEBRAS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument in both directions; the only identification it uses is stated in the span, and no step is left to the reader.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Centralizers of subalgebras. Same chapter, the definition of the centralizer of a subalgebra in an algebra. Supplies the objects C and C' and the centralizer C" computed in the proof.
- Tensor products of algebras and their subspaces. Same chapter, the elementary description of elements of a tensor product of algebras and of the intersection of the two standard subalgebras. Supplies the identification of the intersection of the tensor product with one factor by the other with the tensor product of the subalgebras.

Lemma 0.1. *Let A, A' be k-algebras. Let $B \subset A$, $B' \subset A'$ be subalgebras with centralizers C, C'. Then the centralizer of $B \otimes_k B'$ in $A \otimes_k A'$ is $C \otimes_k C'$.*

Proof. Denote $C'' \subset A \otimes_k A'$ the centralizer of $B \otimes_k B'$. It is clear that $C \otimes_k C' \subset C''$. Conversely, every element of C'' commutes with $B \otimes 1$ hence is contained in $C \otimes_k A'$. Similarly $C'' \subset A \otimes_k C'$. Thus $C'' \subset C \otimes_k A' \cap A \otimes_k C' = C \otimes_k C'$. □

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